# Week 4 – Cyber Security Internship Project Report

Virtual Lab Setup, Web Server Deployment,
Network Scanning and Traffic Analysis

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# Table of Contents

1. Introduction
    1.1 Background  1.2 Introduction to Network Security  1.3 Importance of Network Security
2. Project Objectives
3. Theoretical Background
    3.1 Computer Networks – 3.14 DNS
4. Lab Environment
5. Web Server Configuration
6. Nmap Network Scanning
7. Wireshark Traffic Analysis
8. Security Observations
9. Complete Workflow
10. Limitations
11. Learning Outcomes
12. Conclusion

References

# 1. Introduction

## 1.1 Background

In the contemporary digital landscape, cybersecurity has emerged as a critical discipline safeguarding information systems, networks, and data from unauthorized access, theft, and damage. As organizations increasingly depend on interconnected systems for operations, the demand for skilled cybersecurity professionals capable of understanding network architectures, identifying vulnerabilities, and implementing defensive measures has grown substantially.

Network security forms the foundation of cybersecurity practices, encompassing the policies, technologies, and procedures designed to protect network infrastructure and the data transmitted across it. Understanding network behavior, service configurations, and traffic patterns is essential for security professionals tasked with defending organizational assets against evolving threats.

## 1.2 Introduction to Network Security

Network security refers to the comprehensive set of measures implemented to protect the integrity, confidentiality, and availability of computer networks and data. It involves both hardware and software technologies, administrative controls, and procedural safeguards designed to prevent unauthorized access, misuse, modification, or denial of network resources.

Key objectives of network security include:

- **Confidentiality:** Ensuring that information is accessible only to authorized individuals
- **Integrity:** Maintaining the accuracy and completeness of data during transmission and storage
- **Availability:** Ensuring that network resources and services are accessible when needed
- **Authentication:** Verifying the identity of users and systems accessing the network
- **Authorization:** Controlling access rights and permissions for authenticated entities

## 1.3 Importance of Network Security

The significance of network security in modern organizations cannot be overstated. With the proliferation of cyber threats including malware, ransomware, phishing attacks, and advanced persistent threats (APTs), robust network security measures are essential for:

- **Protecting Sensitive Data:** Preventing unauthorized access to confidential information such as customer data, intellectual property, and financial records.
- **Maintaining Business Continuity:** Ensuring that critical network services remain operational and available to support business operations.
- **Regulatory Compliance:** Meeting legal and regulatory requirements for data protection (e.g., GDPR, HIPAA, PCI-DSS).
- **Reputation Management:** Avoiding security breaches that could damage organizational reputation and customer trust.

- **Cost Avoidance:** Preventing financial losses associated with data breaches, system downtime, and recovery efforts.

Network security testing, when performed ethically and within authorized environments, enables organizations to identify weaknesses before malicious actors can exploit them, forming a cornerstone of proactive defense strategies.

# 2. Project Objectives

The primary objectives of this Week 4 cybersecurity internship task were:

- **Network Setup:** Establish a functional virtual network with minimum two machines (attacker and target) using VMware virtualization software, ensuring proper IP configuration and network connectivity.
- **Service Configuration:** Install and configure a web server (Apache HTTP Server) on the target machine and verify operational status through service checks and accessibility tests.
- **Network Scanning:** Utilize Nmap (Network Mapper) to perform network reconnaissance on the target machine, identify open ports, enumerate running services, and document findings systematically.
- **Traffic Analysis:** Capture and analyze network traffic using Wireshark, focusing on ICMP, HTTP, and DNS protocols to understand packet structures and communication patterns.
- **Workflow Documentation:** Demonstrate complete workflow from network setup through service deployment, scanning, traffic analysis, and final findings in a structured manner.
- **Security Understanding:** Develop practical comprehension of network reconnaissance techniques, service enumeration, packet analysis, and their defensive implications for network security.
- **Ethical Practice:** Reinforce the importance of conducting security testing only within authorized, isolated environments, adhering to ethical guidelines and legal frameworks governing cybersecurity research.

# 3. Theoretical Background

## 3.1 Computer Networks

A computer network is a collection of interconnected computing devices capable of communicating and sharing resources with each other. Networks enable data exchange between devices using standardized communication protocols, facilitating resource sharing, communication, and distributed computing.

- **LAN (Local Area Network):** Covers a small geographic area such as a home, office, or building
- **MAN (Metropolitan Area Network):** Spans a city or metropolitan area
- **WAN (Wide Area Network):** Extends over large geographic areas, often connecting multiple LANs

## 3.2 Network Security

Network security encompasses the policies, practices, and technologies implemented to protect network infrastructure, data, and resources from unauthorized access, attacks, and misuse. It involves multiple layers

of defense including physical security (protecting hardware and infrastructure), technical security (firewalls, encryption, access controls), and administrative security (policies, procedures, user training).

## 3.3 IP Addressing

An IP (Internet Protocol) address is a numerical label assigned to each device connected to a network that uses the Internet Protocol for communication. IPv4 uses 32-bit addresses, typically expressed in dotted-decimal notation (e.g., 192.168.86.136), providing approximately 4.3 billion unique addresses.

A subnet mask is a 32-bit number that divides an IP address into network and host portions. Common subnet masks include 255.255.255.0 (/24), which allows 254 host addresses. Private IP ranges include 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16. The lab network (192.168.86.0/24) falls within the private IP address range.

## 3.4 TCP/IP

TCP/IP is the fundamental communication protocol suite used for interconnecting network devices. TCP (Transmission Control Protocol) is a connection-oriented protocol that provides reliable, ordered, and error-checked delivery of data, establishing a connection through a three-way handshake (SYN, SYN-ACK, ACK).

## 3.5 Ports and Services

A port is a logical endpoint for communication, identified by a 16-bit number (0–65535). Well-known ports (0–1023) are reserved for system services (HTTP: 80, HTTPS: 443, SSH: 22); registered ports (1024–49151) are assigned to user applications; dynamic/private ports (49152–65535) are used for temporary connections. Port 80 is the standard TCP port for unencrypted HTTP traffic, while port 443 is used for encrypted HTTPS traffic.

## 3.6 Web Servers

A web server accepts HTTP requests from clients and serves web content in response. Apache HTTP Server is an open-source web server maintained by the Apache Software Foundation, widely used for its reliability, flexibility, and extensive module support (authentication, URL rewriting, SSL/TLS).

## 3.7 HTTP and HTTPS

HTTP is an application-layer, request-response protocol that is stateless and unencrypted, meaning transmitted data can be intercepted and read by anyone with network access. HTTPS extends HTTP with TLS/SSL encryption, protecting data integrity and confidentiality. Modern best practice mandates HTTPS for all web communications, especially those involving sensitive data.

## 3.8 Network Reconnaissance

Network reconnaissance is the process of gathering information about a target network — active hosts, open ports, running services, and topology. It is typically the first phase of a security assessment and includes host discovery, port scanning, service enumeration, and OS detection.

## 3.9 Nmap

Nmap (Network Mapper) is a free, open-source utility for network discovery and security auditing, developed by Gordon Lyon (Fyodor). It uses raw IP packets to determine hosts available on a network, services and versions offered, operating systems in use, and the type of packet filters/firewalls present. Nmap supports

TCP SYN, TCP connect, UDP, and specialized scans (FIN, NULL, Xmas), plus the Nmap Scripting Engine (NSE) for advanced detection.

## 3.10 Packet Analysis

Packet analysis (protocol analysis / network forensics) is the process of capturing, inspecting, and interpreting network traffic. A packet contains a payload and headers (source/destination addresses, protocol type, sequence numbers); a frame is the data-link-layer representation including MAC addressing and error-checking.

## 3.11 Wireshark

Wireshark is a widely-used, open-source network protocol analyzer that captures and interactively browses network traffic, supporting deep inspection of hundreds of protocols, live capture and offline analysis, and export to formats such as PCAP, XML, and CSV.

## 3.12 ICMP

ICMP (Internet Control Message Protocol) is a network-layer protocol used for diagnostics and control.

Common message types include Echo Request (Type 8) and Echo Reply (Type 0), used by ping; Destination Unreachable (Type 3); and Time Exceeded (Type 11).

## 3.13 HTTP

HTTP operates on a request–response model: the client sends a request (e.g., GET /index.html), the server processes it, and returns a response with a status code and content. Common status codes include 200 OK, 301 Moved Permanently, 403 Forbidden, 404 Not Found, and 500 Internal Server Error.

## 3.14 DNS

DNS (Domain Name System) is a hierarchical, distributed naming system that translates human-readable domain names (e.g., google.com) into IP addresses. DNS operates primarily over UDP port 53, involving DNS queries, responses, resolvers, and records (A, AAAA, MX, CNAME, etc.).

# 4. Lab Environment

## 4.1 Virtualization Environment

VMware Workstation was used as the virtualization platform for this exercise. Virtualization enables multiple operating systems to run concurrently on a single physical host, each isolated from the others and from the host system.

- **Isolation:** VMs operate in sandboxed environments, preventing impact on production systems
- **Snapshot Capability:** VM states can be saved and restored, enabling safe experimentation
- **Resource Efficiency:** Multiple systems can be tested on a single physical machine
- **Network Flexibility:** Virtual networks can be configured in various topologies (NAT, bridged, host-only)

## 4.2 Kali Linux

Kali Linux is a Debian-based distribution designed for digital forensics and penetration testing, pre-installed with tools such as Nmap, Wireshark, Metasploit, and Burp Suite. In this lab it served as the attacker/scanner machine, configured with IP address 192.168.86.137 on interface eth0.

## 4.3 Ubuntu

Ubuntu Linux is a popular Debian-based distribution known for stability and community support. In this lab it served as the target/web server machine hosting Apache HTTP Server, configured with IP address 192.168.86.136 on interface ens33.

## 4.4 Network Configuration

Both virtual machines were configured within the same virtual network to enable direct communication:

| Machine | Operating System | IP Address | Network Interface | Role |
|---|---|---|---|---|
| Kali | Kali Linux | 192.168.86.137 | eth0 | Attacker / Scanner |
| Target | Ubuntu Linux | 192.168.86.136 | ens33 | Web Server |

*Table 1: Lab Machine Configuration*

Network Subnet: 192.168.86.0/24 | Subnet Mask: 255.255.255.0 | Network Mode: NAT (isolated from external networks)

## 4.5 Network Architecture

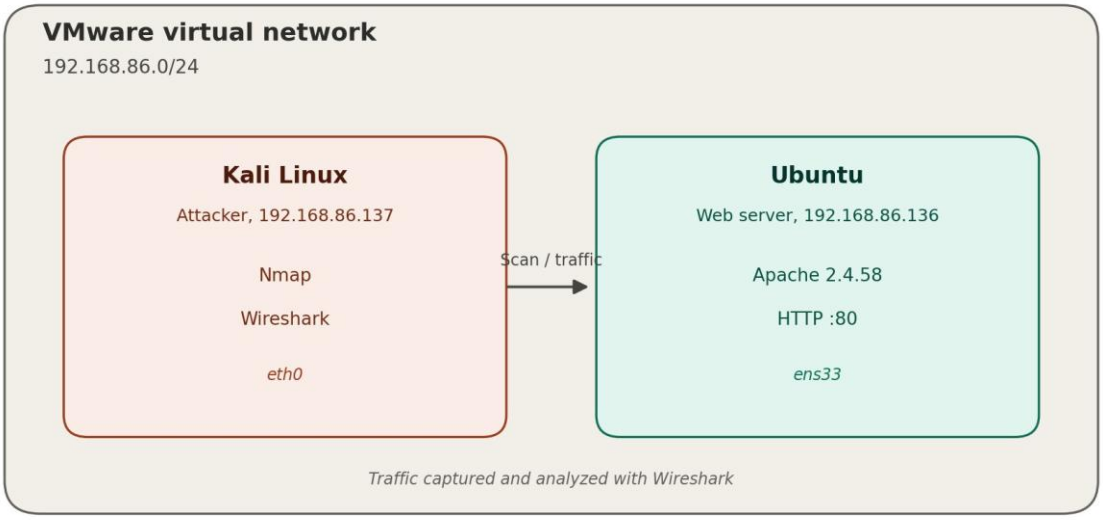



*Figure 1: Virtual lab network architecture – Kali Linux (attacker/scanner, 192.168.86.137) and Ubuntu (target/web server, 192.168.86.136) on the 192.168.86.0/24 VMware virtual network.*

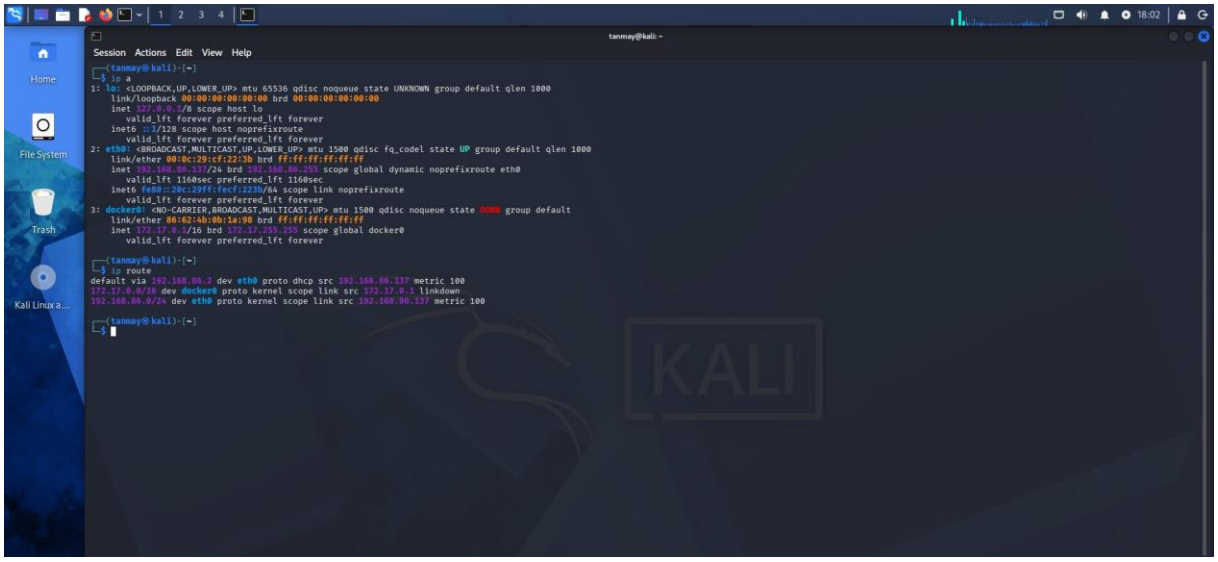



*Figure 2: Kali Linux IP configuration showing 192.168.86.137 on interface eth0.*

```
tanmay@tanmay-VMware-Virtual-Platform:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:a7:50:7f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.86.136/24 brd 192.168.86.255 scope global dynamic noprefixroute
 ens33
       valid_lft 1515sec preferred_lft 1515sec
    inet6 fe80::20c:29ff:fea7:507f/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOW
N group default
    link/ether 6a:ac:94:3f:da:f5 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
tanmay@tanmay-VMware-Virtual-Platform:~$
```

*Figure 3: Ubuntu Linux IP configuration showing 192.168.86.136 on interface ens33.*

## 4.6 Connectivity Testing

ICMP ping tests were conducted from Kali Linux to verify network connectivity between the two machines:

```
ping -c 4 192.168.86.136
```

**Results:** Target (192.168.86.136) responded successfully to all ICMP echo requests, with 0% packet loss and a round-trip time under 1 ms — consistent with a local virtual network.

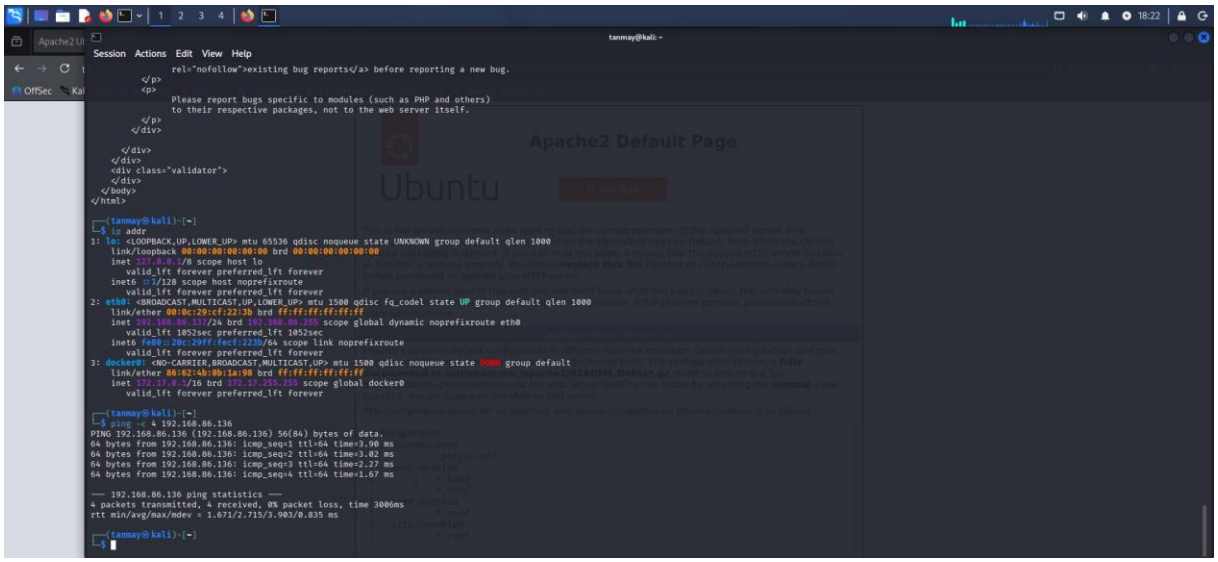



*Figure 4: Successful ping connectivity test from Kali Linux (192.168.86.137) to Ubuntu (192.168.86.136) showing 0% packet loss.*

Connectivity verification confirmed that both machines were properly configured within the same network segment and could communicate with each other, establishing the foundation for subsequent security testing activities.

# 5. Web Server Configuration

## 5.1 Apache Installation

Apache HTTP Server was installed on the Ubuntu target machine to simulate a web service for security testing — one of the most widely deployed web servers globally, making it a realistic target for assessment exercises.

```
sudo apt update sudo apt
install apache2 -y sudo
systemctl start apache2 sudo
systemctl enable apache2
```

- sudo apt update — updates package repository information
- sudo apt install apache2 -y — installs Apache HTTP Server with automatic confirmation
- sudo systemctl start apache2 — starts the Apache service immediately
- sudo systemctl enable apache2 — configures Apache to start automatically on boot

## 5.2 Apache Service Verification

Service status was verified using systemctl to ensure Apache was running correctly:

```
sudo systemctl status apache2
```

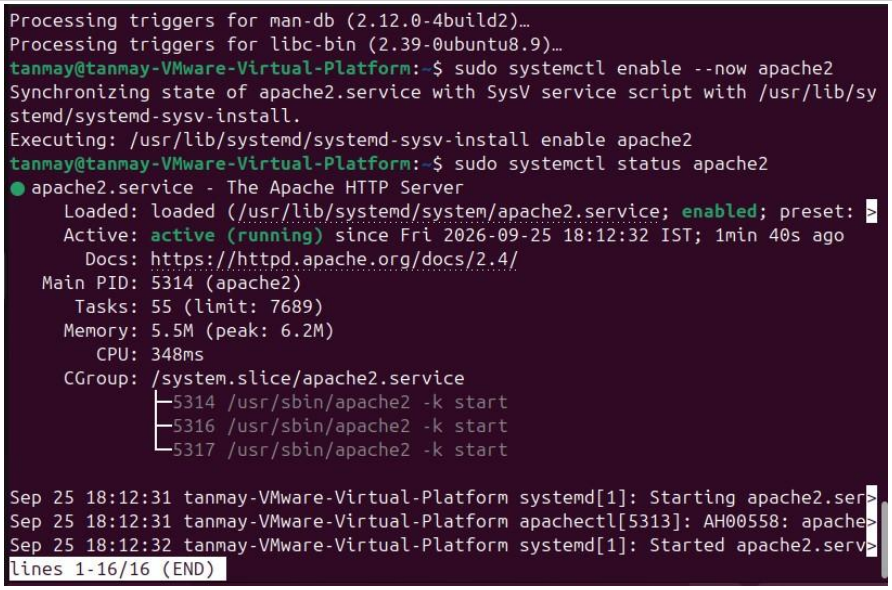



*Figure 5: Apache service status showing active (running) state on the Ubuntu target machine.*

Successful service status confirmation indicated that Apache was operational and listening for incoming HTTP requests on TCP port 80.

## 5.3 Webpage Verification

The Apache default webpage was accessed from Kali Linux to verify that the web service was reachable across the network, via curl http://192.168.86.136 and directly through the browser at http://192.168.86.136. The default Apache2 Ubuntu default page with the message “It works!” was displayed, confirming successful deployment and network accessibility.

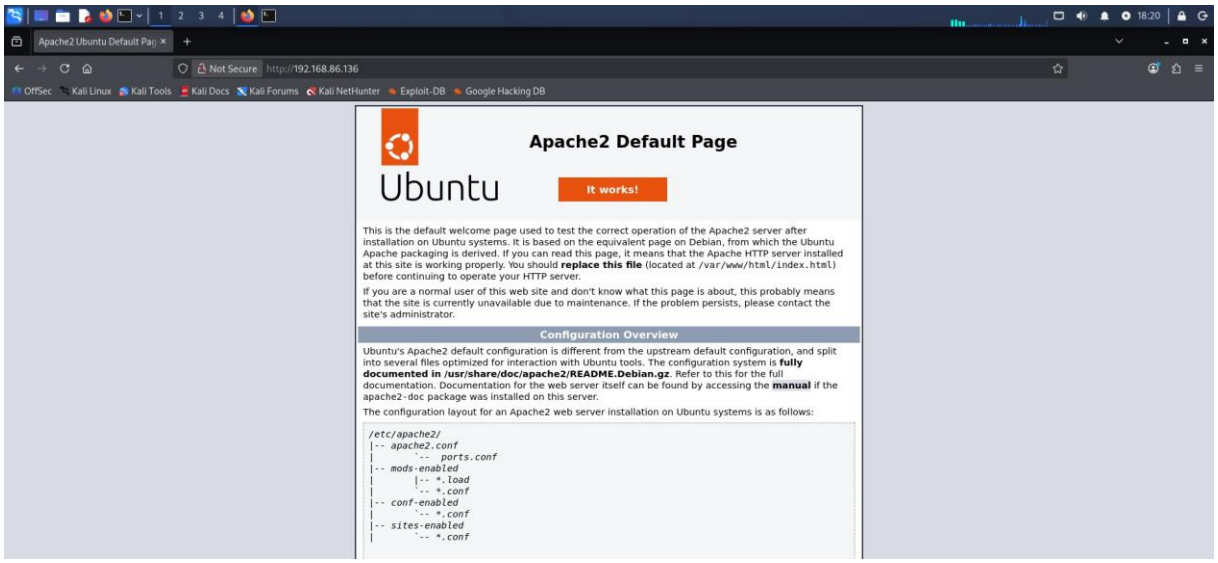



*Figure 6: Apache2 Ubuntu default page accessed from the Kali Linux browser, confirming web server reachability at http://192.168.86.136.*

Successful access to the Apache default webpage from the Kali machine validated that Apache HTTP Server was properly installed and running, network connectivity between Kali and Ubuntu was functional, TCP port 80 was open and accepting HTTP connections, and the web server was correctly configured to serve content.

# 6. Nmap Network Scanning

## 6.1 Nmap Introduction

Nmap is the industry-standard, free and open-source utility for network discovery and security auditing. It uses raw IP packets to determine hosts available on a network, services and versions in use, operating systems, and firewall/packet-filter configurations.

## 6.2 Basic Scan

```
nmap 192.168.86.136
```

**Purpose:** performs a default TCP SYN scan of the top 1,000 most common ports. Without options, Nmap performs a TCP SYN scan (-sS) by default, reporting each port as **open** (an application is actively listening), **closed** (reachable but nothing listening), or **filtered** (blocked by a firewall or other obstacle).

## 6.3 Service Version Detection

```
nmap -sV 192.168.86.136
```

**Purpose:** the -sV flag enables version detection, sending probes to open ports to elicit responses that are compared against Nmap's fingerprint database, revealing application name, version, and sometimes OS hints — more accurate than port-based guessing alone.

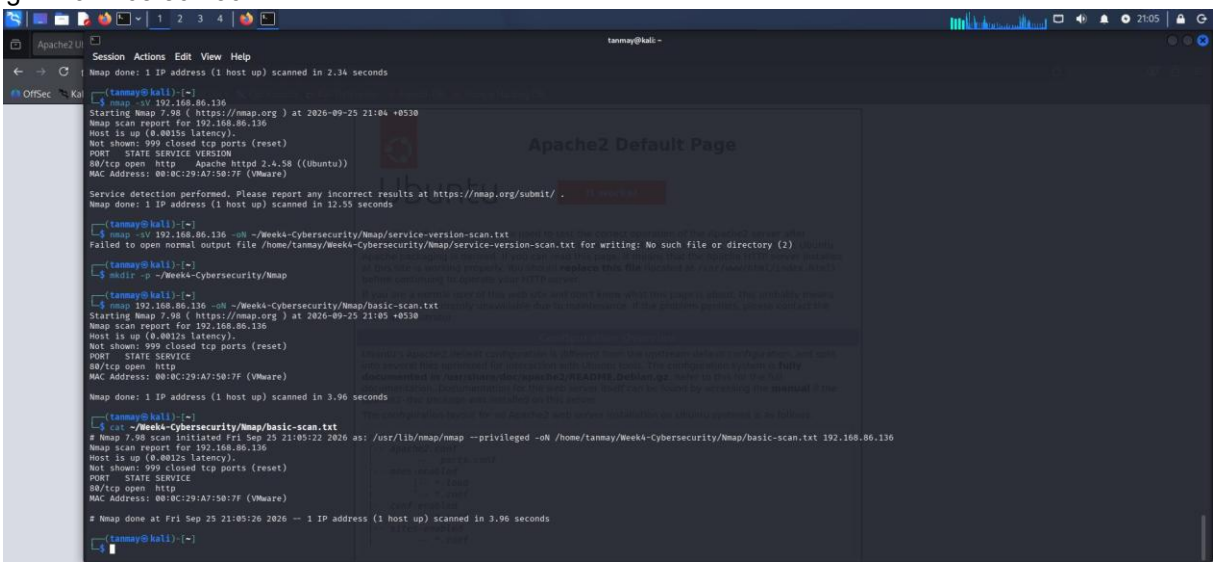
## 6.4 Default Script Scan

```
sudo nmap -sC -sV 192.168.86.136
```

**Purpose:** runs default Nmap Scripting Engine (NSE) scripts alongside version detection. -sC runs the default (safe) script category for non-intrusive enumeration; -sV adds version detection; NSE scripts can extract HTTP titles, server headers, and other service-specific details.

## 6.5 Nmap Results

Target: 192.168.86.136




*Figures 7 & 8: Nmap basic scan and service-version detection, run and saved to file on Kali (basic-scan.txt, service-version-scan.txt), showing TCP port 80 open running Apache httpd 2.4.58 (Ubuntu).*

### *Basic scan output (basic-scan.txt)*

```
# Nmap 7.98 scan initiated Fri Sep 25 21:05:22 2026 as: /usr/lib/nmap/nmap --privileged -oN
/home/tanmay/Week4-Cybersecurity/Nmap/basic-scan.txt 192.168.86.136
Nmap scan report for 192.168.86.136
Host is up (0.0012s latency).
Not shown: 999 closed tcp ports (reset)
PORT STATE SERVICE
80/tcp open http
MAC Address: 00:0C:29:A7:50:7F (VMware)

# Nmap done at Fri Sep 25 21:05:26 2026 -- 1 IP address (1 host up) scanned in 3.96 seconds
```

### *Service version detection output (service-version-scan.txt)*

```
# Nmap 7.98 scan initiated Fri Sep 25 21:06:51 2026 as: /usr/lib/nmap/nmap --privileged -sV -oN
/home/tanmay/Week4-Cybersecurity/Nmap/service-version-scan.txt 192.168.86.136
Nmap scan report for 192.168.86.136
Host is up (0.0011s latency).
Not shown: 999 closed tcp ports (reset)
PORT STATE SERVICE VERSION
80/tcp open http Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 00:0C:29:A7:50:7F (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Fri Sep 25 21:07:01 2026 -- 1 IP address (1 host up) scanned in 10.11 seconds
```

### *Default script scan output (default-script-scan.txt)*

```
# Nmap 7.98 scan initiated Fri Sep 25 21:08:53 2026 as: /usr/lib/nmap/nmap -sC -sV -oN
/home/tanmay/Week4-Cybersecurity/Nmap/default-script-scan.txt 192.168.86.136
Nmap scan report for 192.168.86.136
Host is up (0.0010s latency).
Not shown: 999 closed tcp ports (reset)
PORT STATE SERVICE VERSION
80/tcp open http Apache httpd 2.4.58 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
MAC Address: 00:0C:29:A7:50:7F (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Fri Sep 25 21:09:06 2026 -- 1 IP address (1 host up) scanned in 13.07 seconds
```

Figure 9: Nmap default script scan output showing HTTP title “Apache2 Ubuntu Default Page: It works” and server header “Apache/2.4.58 (Ubuntu)”.

| Port | Protocol | State | Service | Version | Observation |
|---|---|---|---|---|---|
| 80/tcp | TCP | Open | HTTP | Apache httpd 2.4.58 (Ubuntu) | Web server reachable, default page served |

*Table 3: Nmap findings summary*

## 6.6 Analysis

- **Host Reachability:** the target (192.168.86.136) was confirmed up and reachable with low latency (0.0011–0.0013s), consistent with a local virtual network.
- **Open Port Identification:** TCP port 80 was identified as open, indicating an active listening service.
- **Service Enumeration:** the service on port 80 was identified as Apache httpd 2.4.58 on Ubuntu Linux.
- **Additional Information:** default NSE scripts extracted the HTTP page title and server header, confirming the web server configuration.

- **Closed Ports:** 999 other scanned TCP ports were reported closed, indicating no other common services were running.

**Important note:** an open port indicates a listening and reachable service but does not by itself prove a vulnerability. Identifying Apache 2.4.58 enables further research but does not constitute evidence of exploitable weaknesses — vulnerability assessment would require additional testing beyond the scope of this exercise.

# 7. Wireshark Traffic Analysis

## 7.1 Wireshark Introduction

Wireshark enables deep inspection of network traffic through live packet capture and interactive analysis, supporting hundreds of protocols, powerful filtering and colorization, and export to formats such as PCAP, XML, and CSV.

## 7.2 Packet Capture

- **Interface:** eth0 (Kali Linux network interface)
- **Capture file format:** PCAPNG (Packet Capture Next Generation)
- **File name:** Week4-Traffic-Capture.pcapng

Traffic was generated through targeted activities: ICMP (ping -c 5 192.168.86.136), HTTP (browser access to http://192.168.86.136), and DNS (nslookup google.com).

## 7.3 ICMP Analysis

**Filter:** icmp. ICMP Echo Request packets were captured travelling from 192.168.86.137 (Kali) to 192.168.86.136 (Ubuntu), with Echo Reply packets returning; sequence numbers incremented with each ping and no packet loss was observed, consistent with the successful ping test.

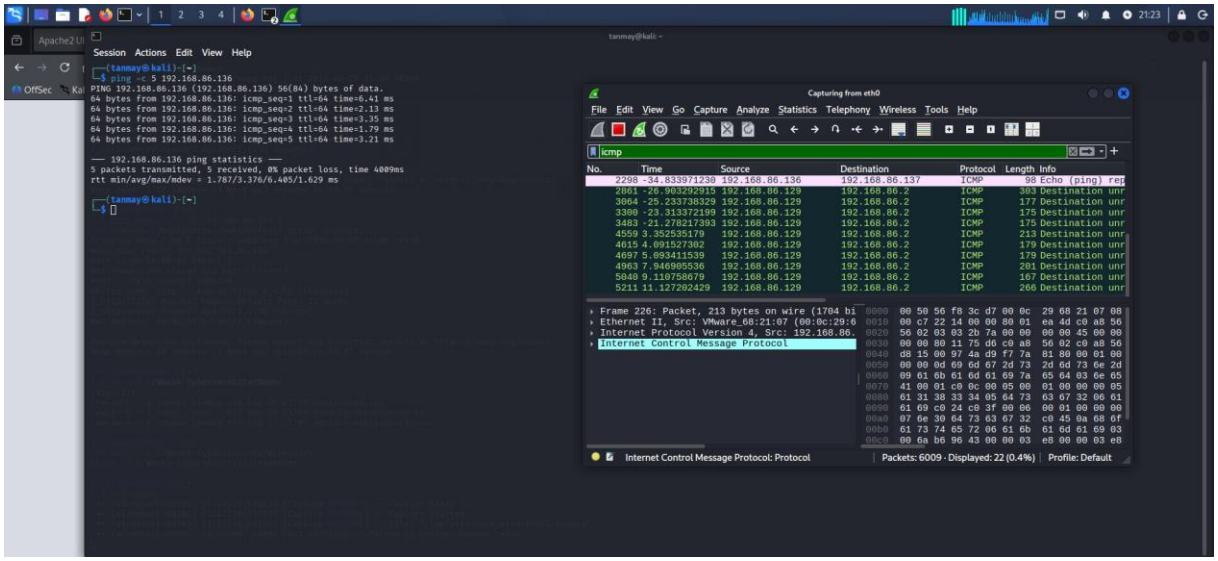



*Figure 10a: Wireshark ICMP capture showing Echo Request/Reply traffic and a decoded ICMP frame.*

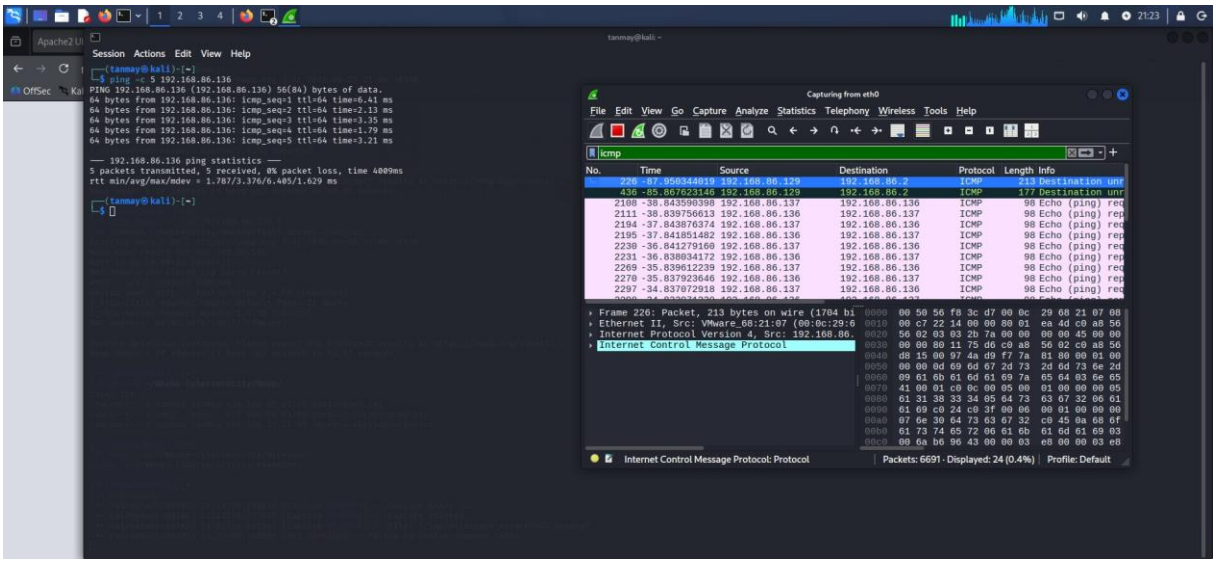


*Figure 10b: Wireshark ICMP capture filtered to Echo Request (Type 8) and Echo Reply (Type 0) packets between Kali (192.168.86.137) and Ubuntu (192.168.86.136).*

## 7.4 HTTP Analysis

**Filter:** http. An HTTP GET request was captured from the Kali browser to the target web server (GET / HTTP/1.1, Host: 192.168.86.136), with a response of HTTP/1.1 200 OK, server header Apache/2.4.58 (Ubuntu), and content type text/html; charset=UTF-8.

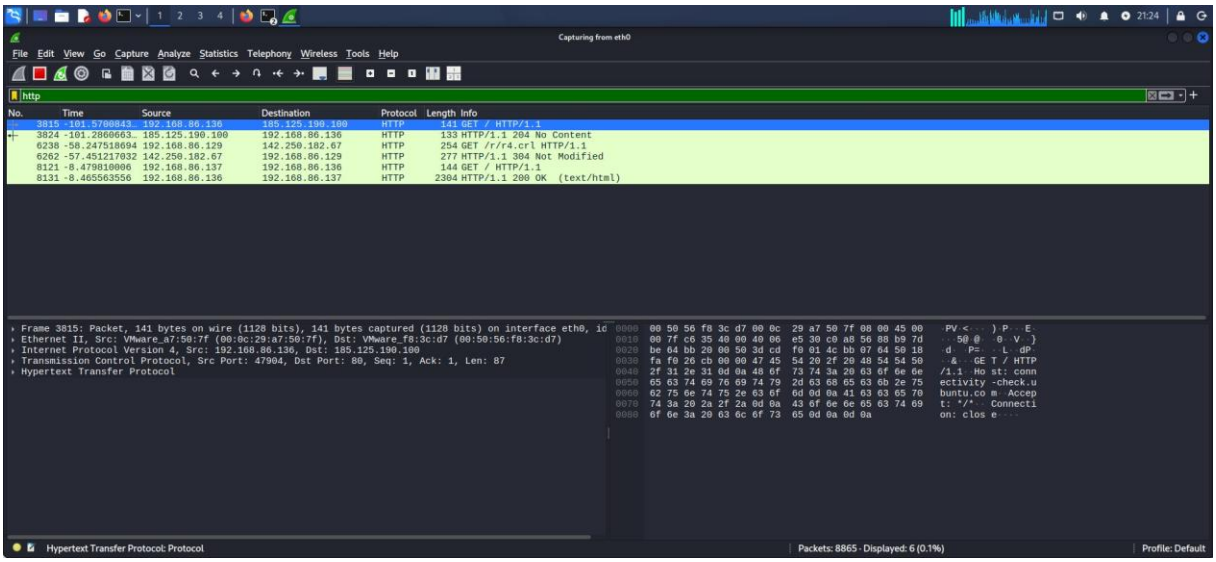


*Figure 11: Wireshark HTTP traffic capture showing the GET request from Kali and the HTTP 200 OK response from the Ubuntu web server.*

**Security consideration:** this HTTP traffic was transmitted in plaintext, allowing full inspection of requests and responses. Production environments should use HTTPS to encrypt sensitive web communications and prevent eavesdropping.

## 7.5 DNS Analysis

**Filter:** dns. DNS query packets were captured travelling from Kali Linux to its configured DNS resolver (query type A / IPv4), with response packets returning resolved IP addresses over standard DNS port 53/UDP.

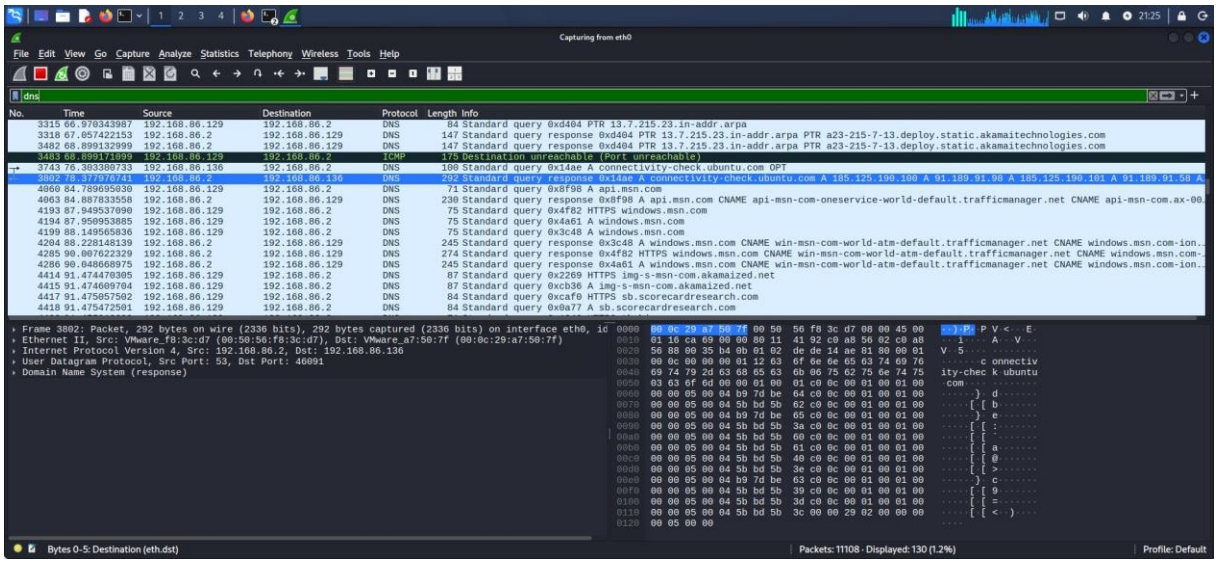


*Figure 12: Wireshark DNS traffic capture showing DNS query and response packets for domain name resolution.*

*Note:* the DNS traffic observed reflects name-resolution activity initiated from the Kali machine during the capture session and does not indicate that the Ubuntu target was running a DNS service.

## 7.6 PCAPNG File

Capture file: **Week4-Traffic-Capture.pcapng**. PCAP/PCAPNG files preserve complete packet data (headers and payloads) for offline analysis, sharing, and archival. Because capture files can contain sensitive information — IP addresses and topology, unencrypted application data, credentials sent in plaintext, and internal hostnames — they should be stored securely, handled carefully, sanitized before sharing, and deleted when no longer needed.

| Protocol | Filter | Purpose | Observation |
|---|---|---|---|
| ICMP | icmp | Connectivity verification | Echo request/reply traffic observed between Kali and Ubunt |
| HTTP | http | Web communication | HTTP GET request and 200 OK response observed |
| DNS | dns | Name resolution | DNS query/response traffic observed for domain resolution |

*Table 4: Wireshark protocol analysis summary*

# 8. Security Observations

## 8.1 Open Service Exposure

The Nmap scan revealed TCP port 80 as open, indicating that Apache was actively listening and accepting connections. An open port represents a potential entry point and increases the attack surface — the total number of points where an unauthorized user could attempt to enter or extract data. In production, every open port should be justified by business need, and unnecessary services should be disabled.

## 8.2 Service Identification

Nmap's version detection identified the web server as Apache httpd 2.4.58 on Ubuntu Linux. This disclosure, while normal for many web servers, enables potential attackers to research known vulnerabilities for that version, tailor attacks accordingly, and infer the underlying operating system for further reconnaissance.

Server-token hardening can mitigate this exposure.

## 8.3 HTTP Security Considerations

Traffic captured in this exercise was unencrypted HTTP, fully inspectable through Wireshark. This illustrates key HTTP vulnerabilities:

- **Eavesdropping:** all transmitted data can be read by anyone with network access
- **Data Tampering:** content can be modified in transit without detection
- **Credential Theft:** login credentials sent over HTTP can be intercepted
- **Session Hijacking:** session cookies can be stolen, enabling account takeover

**Recommendation:** production web services handling sensitive data should implement HTTPS to encrypt communications and ensure confidentiality and integrity.

## 8.4 Network Monitoring

The Wireshark capture demonstrated the value of network monitoring: analyzing traffic lets security professionals identify unexpected or unauthorized services, detect anomalous patterns indicative of attacks or misconfiguration, investigate incidents through packet-level forensics, and validate that controls such as firewalls and encryption are functioning correctly.

## 8.5 Security Recommendations

- **Minimize Attack Surface:** disable unnecessary services and close unused ports.
- **Implement Firewall Rules:** configure host-based firewalls (e.g., ufw, iptables) to restrict access to required ports and sources.
- **Use HTTPS:** deploy TLS/SSL certificates to encrypt communications and protect data in transit.
- **Limit Information Disclosure:** minimize version information in HTTP headers and error pages.
- **Regular Scanning:** perform periodic network scans to identify unauthorized services or misconfigurations.
- **Network Segmentation:** isolate critical services to limit lateral movement in case of compromise.

- **Monitoring and Logging:** implement continuous monitoring and maintain comprehensive logs for incident response.
- **Authorized Testing:** always perform security testing only within authorized environments with explicit permission.

# 9. Complete Workflow

The diagram below summarizes the end-to-end security testing process performed during this Week 4 internship task, from initial network setup through to documented security findings.

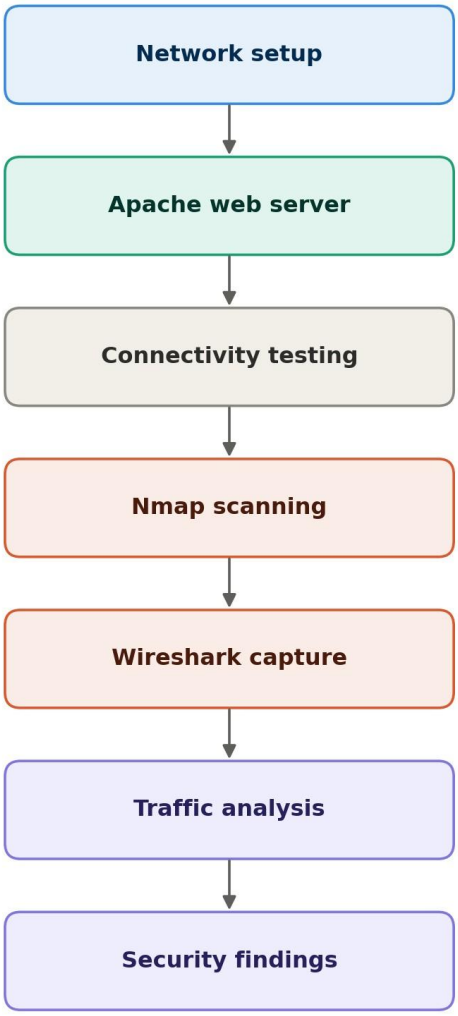



*Figure 14: Week 4 security testing workflow.*

### *Workflow explanation*

- **Network setup:** VMware virtualization was used to create isolated VMs for Kali Linux and Ubuntu Linux.
- **IP configuration:** both machines were assigned addresses (192.168.86.137 and 192.168.86.136) within 192.168.86.0/24 to enable direct communication.
- **Connectivity testing:** ICMP ping tests verified reachability between Kali and Ubuntu.
- **Apache web server:** Apache HTTP Server was installed on Ubuntu to simulate a web service, and its status was verified with systemctl.

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- **Webpage check:** browser access from Kali confirmed the web server was serving content correctly.

  **Nmap scanning:** basic, service-version, and default-script scans were run to enumerate open ports and services.

- **Wireshark capture:** traffic was captured on the Kali interface during ICMP, HTTP, and DNS activity.
- **Traffic analysis:** captured packets were filtered and analyzed to understand protocol behavior.
- **Security findings:** observations were compiled into security implications and recommendations, supported by screenshots and the saved PCAPNG file.

# 10. Limitations

- **Controlled Environment:** the assessment was performed exclusively within an isolated VMware lab, not on production or external networks.
- **Single Target Scope:** only the Ubuntu target machine (192.168.86.136) was tested.
- **Basic Assessment Focus:** the exercise covered network setup, service discovery, and packet analysis — not exploitation, privilege escalation, or post-exploitation.
- **Non-Destructive Testing:** limited to reconnaissance and enumeration; no exploitation or destructive testing was conducted.
- **Point-in-Time Snapshot:** results reflect the lab configuration at the time of testing.
- **Limited Protocol Analysis:** only ICMP, HTTP, and DNS were analyzed in detail.
- **No Firewall Configuration:** the target had no firewall rules, which would affect results in production environments.
- **DNS Traffic Context:** DNS traffic observed reflects Kali's own name resolution, not a DNS service on the target.
- **Version Detection Limitations:** identifying Apache 2.4.58 does not by itself confirm any vulnerability.
- **Short Capture Duration:** the Wireshark captures were brief snapshots rather than extended monitoring.

# 11. Learning Outcomes

### *Technical skills developed*

- Virtual network configuration using VMware, including adapters, IP addressing, and connectivity verification
- IP addressing and subnetting — IPv4 addressing, subnet masks, private IP ranges, segmentation
- Network connectivity testing with ICMP ping
- Web server deployment — installing, configuring, and verifying Apache HTTP Server on Ubuntu
- Service management with systemctl (start, stop, enable, status)
- Nmap scanning — basic scans, service version detection, and NSE script execution
- Port and service identification and interpretation of Nmap output
- Wireshark packet capture — live capture, interface selection, capture file management
- Protocol analysis — filtering and analyzing ICMP, HTTP, and DNS traffic
- Evidence collection through screenshots, capture files, and structured reporting

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### *Security concepts reinforced*

- Attack surface awareness — how open ports and services increase exposure
- Information disclosure through service enumeration
- The importance of encryption, observed firsthand via unencrypted HTTP traffic

  The value of network monitoring for visibility into behavior and potential issues
- Ethical practice — testing only within authorized, isolated environments
- Security documentation — translating technical findings into clear, professional reports

### *Professional development*

- Systematic, structured approach to security assessment from planning through reporting
- Attention to detail in documenting configurations, commands, and results
- Critical thinking — distinguishing observations from conclusions
- Communication skills — explaining technical concepts clearly in written form

# 12. Conclusion

This Week 4 cybersecurity internship task successfully demonstrated fundamental network security assessment techniques within an authorized, isolated virtual lab environment. A functional two-machine network was established using VMware virtualization, with Kali Linux (192.168.86.137) serving as the security testing platform and Ubuntu Linux (192.168.86.136) hosting an Apache HTTP Server.

The Apache web server (version 2.4.58) was successfully deployed and verified for accessibility through both service status checks and browser-based testing. Nmap network scanning identified TCP port 80 as open with HTTP service running, and version detection accurately enumerated the Apache software and version. Default NSE scripts provided additional information including the HTTP page title and server header.

Wireshark packet capture enabled deep inspection of network traffic, successfully capturing and analyzing ICMP (Echo Request/Reply), HTTP (GET request and 200 OK response), and DNS (query/response) protocols. The unencrypted nature of HTTP traffic was clearly demonstrated, reinforcing the importance of TLS/SSL encryption for sensitive web communications.

Key security observations included the relationship between open ports and attack surface, the implications of service version disclosure, the risks of unencrypted communications, and the value of network monitoring — informing practical security recommendations applicable to production environments.

This exercise provided invaluable hands-on experience bridging theoretical knowledge with practical implementation. The skills developed in virtual network configuration, service deployment, network reconnaissance, and traffic analysis form the foundation for advanced cybersecurity competencies essential for professional practice in network defense, penetration testing, and security operations.

The importance of conducting security testing ethically and within authorized environments was reinforced throughout this exercise. Future work could expand upon this foundation by implementing firewall configurations, deploying HTTPS with TLS certificates, performing vulnerability assessments with NSE scripts, and analyzing additional protocols and attack vectors.